

Fuzzy logic, genetic algorithms, and artificial neural networks applied to cognitive radio networks: A review

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Abstract

Cognitive radios are expected to play an important role in capturing the constantly growing traffic interest on remote networks. To improve the usage of the radio range, a cognitive radio hub detects the weather, evaluates the open-air qualities, and then makes certain decisions and distributes the executives' space assets. The cognitive radio works in tandem with artificial intelligence and artificial intelligence methodologies to provide a flexible and intelligent allocation for continuous production cycles. The purpose is to provide a single source of information in the form of a survey research to enable academics better understand how artificial intelligence methodologies, such as fuzzy logics, genetic algorithms, and artificial neural networks, are used to various cognitive radio systems. The various artificial intelligence approaches used in cognitive radio engines to improve cognition capabilities in cognitive radio networks are examined in this study. Computerized reasoning approaches, such as fuzzy logic, evolutionary algorithms, and artificial neural networks, are used in the writing audit. This topic also covers cognitive radio network implementation and the typical learning challenges that arise in cognitive radio systems.

Keywords

Cognitive radio, fuzzy logics, artificial neural networks, artificial intelligence, genetic algorithm

Date received: 31 January 2022; accepted: 20 June 2022

Handling Editor: Yanjiao Chen

Introduction

According to the Cisco Visual Networking Index, global IP traffic will increase by 168 bytes per month by 2019, with multiple times the global population. Furthermore, assets such as power and data transport speed are limited. As a result, smart adjustments are required to reduce energy consumption while updating asset designation. Joseph Mitola III and Gerald Q Maguire¹ proposed cognitive radio (CR) as a solution for universal range access in 1999. CR is defined as a combination of model-based thinking with radio programming innovation.² In 2005, Simon Haykin³ investigated CR and labeled it as “mind-engaged remote correspondences.”

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CR is a framework that recognizes the climate, analyzes its transmission boundaries, and then makes decisions about dynamic time–recurrence space asset identification and the board to improve the utilization of the radio electromagnetic range.⁴ The radio asset board, in general, wants to increase the use of various radio assets so that the radio framework's exposure is improved. Consider the boundaries of the barrier temperature range; as an example, the inventors of Zhao and Morales-Tirado⁵ introduced an ideal asset (power and transmission capacity) assignment in cognitive radio networks (CRNs), explicitly in the range of fundamental conditions. In some cases, the improvement recipes yield perfect asset assignment arrangements at the expense of global intermingling, calculation time, and complexity. To reduce complexity and achieve feasible continual asset allocations, CRNs should be given learning and reasoning capabilities. The intellectual motor should make the CR's activities easier using artificial intelligence (AI) methods.⁶

CR, according to Gavrilovska,⁷ is a clever remote communication structure that is aware of your current position and uses the agreement to benefit from the climate and establish approaches to accommodate factual variance in upgrading information.

As a result of interacting with its Radio Frequency (RF) environment, a CR should be astute and ready to profit from its interactions. As a result, learning is an important component of CR, which can be delivered using computerized reasoning as AI approaches.⁸ The CR framework standard and its essential assets, traits, and points are discussed in this review. Then we will look at how AI handles the learning cycle, the importance of learning in CRs, and the learning capacity of CRs. The focus then moves to a written examination of the most recent advances in CRs that employ learning methodologies. Several investigations of learning approaches in CR-assignment tasks have been performed, but they lack key components of a comprehensive study on CR frameworks.⁹

Wygłinski et al.¹⁰ provided a brief overview of artificial reasoning methodologies, but their focus was on the sequence of events and layout of CR applications and proving grounds.¹¹ Support learning, game hypothesis, artificial neural networks (ANNs) or neural networks (NNs), support vector machines, and the Markov model were among the learning methods studied. They also talked about the benefits, limitations, and difficulty of implementing these tactics in CR exercises. Game theory, reinforcement learning, Bayesian networks, fuzzy logic (FL), and case-based logic are all investigated by the author of this article.¹² Instead of writing, we present a complete study that considers all learning approaches used in intellectual networks. ANNs, FL, and genetic algorithms (GAs) are among

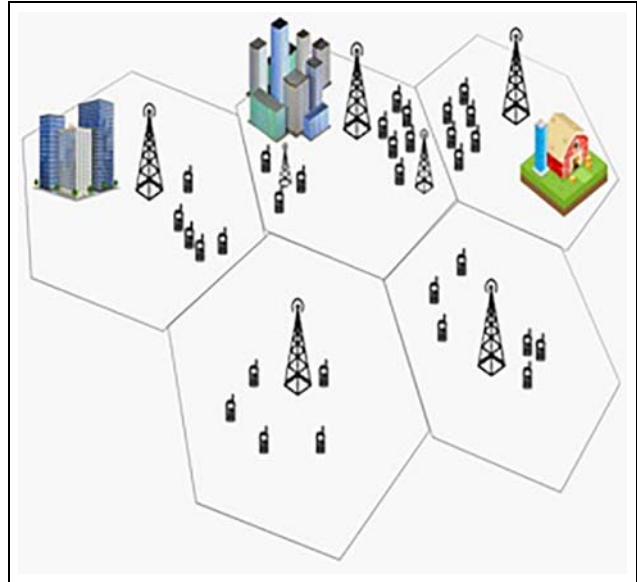


Figure 1. Wireless networking developed by cognitive radio networks.

the automated reasoning techniques used in the summary.¹²

The intellectual cycle is depicted in Figure 1. The remote correspondence framework, as shown in Figure 1, is made up of base stations (BSs) and radio networks, with some serving as fundamental/ primary users (PUs) or networks within its range, and others serving as auxiliary/ secondary users (SUs) who may use it when it is not in use by other networks. CRs are expected to play a key role in meeting the constantly growing traffic demands on wireless networks.¹³ A CR node detects the surroundings, analyzes the outside features, and then makes decisions for dynamic time–frequency–space resource allocation and management to optimize the use of the radio spectrum.¹⁴

The CRN, as shown in Figure 2, followed the intellectual cycle, the board, and network execution for an ideal asset. It begins by identifying the climate, assessing external variables, and then deciding on dynamic asset distribution and executive decisions to improve the usage of radio electromagnetic range.¹⁵ An elemental analysis of them will follow. It should be highlighted that, in this situation, the learning agent is not passive and can not only observe but can also actively change the state of the system through its activities, potentially moving the environment to a desirable condition that rewards the agent with the maximum reward.¹⁶

Environmental sensing

In CRNs, the virtual network requires the auxiliary network for range use. The optional network may use

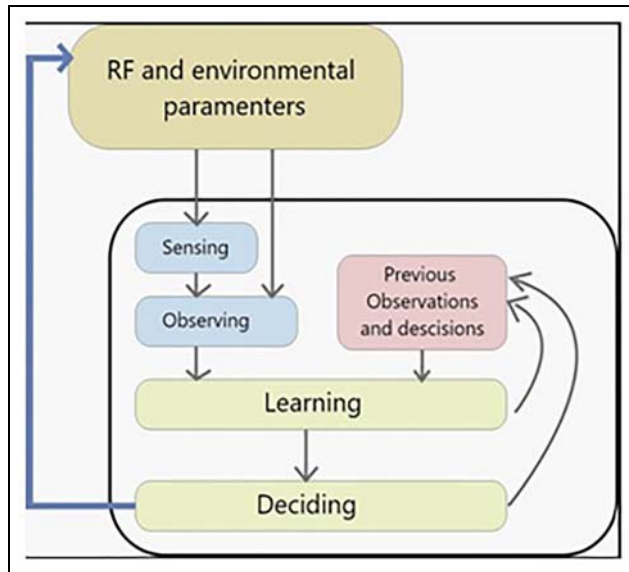


Figure 2. Cognitive radio's learning process.

the available range if it does not interfere with the virtual network. As a result, it should first assess and detect boundaries in a short amount of time, such as (1) channel qualities between the BS and clients; (2) range and power accessibility; (3) range opening accessibility focusing on recurrence, time, and space; (4) client and application requirements; (5) power utilization; and (6) nearby strategies and other imperatives.¹⁷

Dissecting the boundaries of the climate

The discovered climate boundaries will be used as contributions for the executives' asset in all aspects, such as time, recurrence, and space. Some of the major asset component aims in CR are (1) limiting piece blunder rate, (2) limiting power use, (3) limiting obstruction, (4) expanding throughput, (5) working on nature of administration, (6) boosting range productivity, and (7) increasing client nature of involvement. In general,¹⁸ CR seeks to achieve many goals; combining specific goals may result in incompatible arrangements, such as lowering power consumption and spot blunder rate at the same time.

Literature review

The application of CR sensor networks (CRSNs) was described by Suh et al.¹⁹ and Kaur et al.²⁰ Several information systems for elderly care have been developed using cognitive sensor networks to address the dilemma of a super-aging society. However, many sensor-network-based systems that monitor radios, resources, habitats, and other factors are not practical or useful for the elderly. As a result, the goal of this

research is to present novel research methodologies for constructing smart living environments for the elderly using converging information technologies, such as cognitive sensor networks and architectural design. In this article, we review the literature to clarify the concept of smart houses and look at cognitive sensor-network-based systems for smart elder housing. Following that, this research explores research avenues for cognitive wireless sensor networks, not only for geriatric smart home services but also for integrating architectural technologies. The proposed directions are to use CR technologies, to category sensor networking devices according to the types of elderly people, to expand environmental sensing services for elderly housing to a wider area, to integrate sensor network circuits into Building Information Modeling (BIM) systems, and to investigate network architecture that is appropriate for construction projects.

Furthermore, in 2018, Liu et al.²¹ suggested CR as a potential strategy for boosting spectrum utilization by allowing cognitive users (CUs) access to the licensed spectrum when the Primary User (PUs) are unavailable. They developed a resource allocation paradigm for spectrum assignment in CRNs based on graph theory in this article. The concept considers the interference restrictions for principal users and the possibility of CU collision. They created a system utility function based on the provided model to optimize the system benefit. They design an improved ant colony optimization algorithm (IACO) from two perspectives based on the proposed model and objective problem: first, they introduce a differential evolution (DE) process to accelerate convergence speed via a monitoring mechanism; and second, they design a variable neighborhood search (VNS) process to avoid the algorithm falling into the local optimal. According to simulation data, the updated technique performs better.

A CR engine platform is proposed in this study^{22–24} for exploiting available frequency channels for a tactical wireless sensor network while attempting to protect incumbent communication devices, referred to as the principal user (PU), from unintended negative interference. There is a compelling need in tactical communication networks to identify accessible frequencies for opportunistic and dynamic access to channels when the PU is active. This article offers a cognitive engine platform for locating available channels using a case-based reasoning technique, which can be used as a crucial capacity on a CR engine to enable high-fidelity dynamic spectrum access (DSA). To this purpose, a credible learning engine to describe the channel use pattern is developed to extract the optimal channel option for the tactical cognitive radio node (TCRN). Simulation tests were used to assess the performance of the proposed cognitive engine, demonstrating the dependability of the functional element, which includes

the learning engine and the case-based reasoning engine. Furthermore, the TCRN's effectiveness in avoiding contact with the PU operation, as measured by the etiquette SU, was demonstrated.

El Morabit et al.²⁵ provide a comprehensive analysis of various AI methodologies used in CR engines to improve cognition capabilities in CRNs. Learning, thinking, decision-making, self-adaptation, self-organization, and self-stability are only few of the human biological processes that AI can emulate. The employment of AI techniques is investigated in applications related to CR's key objectives, such as spectrum sensing, spectrum sharing, spectrum mobility, and decision-making for issues, such as DSA, resource allocation, parameter modification, and optimization. The goal is to provide a single source as a survey study to help academics better understand the many AI methodologies used in different CR designs and to connect interested readers to current AI research in CRNs.

Guru et al.²⁶ show that in the year 2022, the CRN has the following sensing procedure, in which the selection and selecting of a trustworthy channel from a list of free channels is critical for assignment to CUs for communication with Quality of Service (QoS). In this study, a consistent spectrum selection and decision scheme based on a twofold NN is provided, and its performance is compared to that of GA and back-propagation neural network (BPNN) schemes. The BPNN-adaptive neuro-fuzzy inference system (ANFIS) is a twofold spectrum selection and judging system that combines the BPNN and ANFIS techniques. The factors such as PU status signal intensity, spectrum demand, velocity, and distance are used to select a channel with the required QoS. The simulation analysis shows that the BPNN-ANFIS strategy reduces the risk of blocking and dropping, resulting in an accuracy of more than 92% for reliable channel selection for CU usage. The suggested approach's blocking probability ranges from 1% to 3%, which is significantly lower than the GA (9%–50%) and BPNN (8%–40%). The advised method has a 4% chance of falling, which is lower than the other two plans' 20% chance of falling.

Abd El-Malek et al.²⁷ in the year 2020, evolutionary computation methods will be thoroughly examined in the study of energy harvesting (EH) approaches in multiple-input single-output CRNs (MISO-CRNs). An SU with multiple antennas gathers energy from a hybrid base station (HBS) during the downlink time slot, in addition to harvesting from the PU transmission. As a result, the SU transmits data across Nakagami-m fading channels on the uplink. The SU total transmission power is limited by a certain tolerable PU interference in the underlay paradigm. The HBS has no previous knowledge of the state of the SU battery. For the high signal-to-noise ratio (SNR) area, a closed-form expression for the SU exact outage

probability is derived and simplified to its asymptotic formula. The system average symbol error probability (ASEP) and ergodic capacity have closed-form formulas.²⁸ To reduce the complicated formula of the exact outage probability, the particle swarm optimization (PSO) algorithm is employed to discover the best SU transmission power. The effect of crucial system parameters on total system performance is revealed through simulations and numerical results, which confirm the resulting mathematical analysis. Furthermore, as compared to the standard evenly distributed model, the optimum solution of the power optimization problem demonstrates a significant improvement in system performance.

AI in CR

Experts in AI want machines to tackle tasks in the same way as humans do. The intelligent machine will take note of its surroundings and improve its functionality.²⁹ The most common concerns in AI brainpower include derivation, thinking, critical thinking, information description, and learning.

There are three basic AI techniques: FL, GAs, and NNs, and their mixtures in one domain and three other areas. Each of the technologies has brought successful answers to a wide range of issues in various sectors. Figure 2 depicts the CR AI learning process. They are as follows: (1) detecting radio recurrence (DRR) boundaries, such as channel quality, (2) noticing the climate and dissecting criticism, such as Acknowledgment (ACK) reactions, (3) learning, (4) recording choices and perceptions for refreshing the model and further developing future dynamic exactness, and (5) settling on asset the board issues and changing transmission blunders as needs are.³⁰ Chen et al.³¹ proposed CR standards based on computational thinking and AI. In addition, the developers discussed the expected uses and key concepts that will aid in the development of CR. Some of the learning techniques used in AI are FL, ANNs, GAs, game theory, support vector machines, reinforcement learning, case-based logics, decision-making trees, Bayesian, Entropy, Markov model, multi-agent systems, and artificial bee colony algorithms. The strategies outlined above, however, are the most often used and applied techniques in CRNs.³²

Functions of CRN

The IEEE 802.22 standards are used by the CR to adjust its parameters (such as transmission power, modulation scheme, bandwidth, operating frequency, and so on) in response to environmental variables. The cycle of CRNs is seen in Figure 3. The existence of all four elements is required to establish CRNs.³³

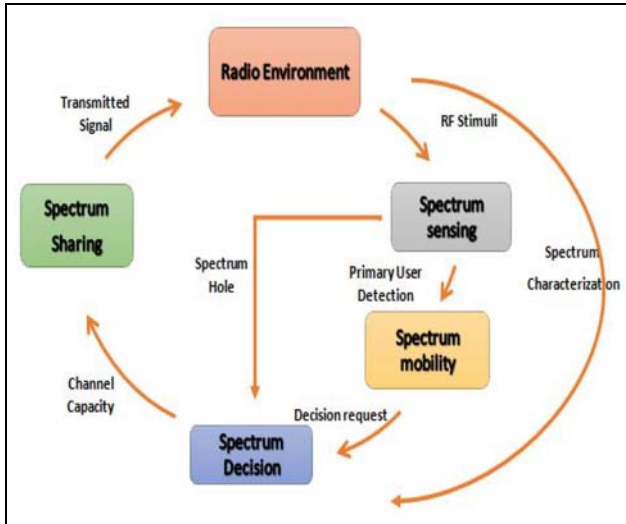


Figure 3. The cycle of cognitive radio networks.

Spectrum sensing

Between PUs and SUs in CRNs, the spectrum sensing network is critical. PUs broadcast data through their licensed spectrum during data transmission. SUs perceive the PU spectrum for data transmission, according to IEEE 802.22 specifications. There are two types of spectrum sensing:

1. Fast sensing, the time frame is 1 ms per channel;
2. Fine sensing is animatedly computed by the BS, depends on the outcomes of the fast sensing, and its work is to detect the frequency band deeply.

The CRs detect the licensed frequency band, and the BS announces whether the licensed band is available or not based on the data given by the CRs to the BS.³³ The energy detection technique is preferred by IEEE 802.22 because it is simple and has a low processing complexity. A number of additional strategies, aside from “Energy Detection,” have been proposed in the writing, for example,

1. Matched filter detection;
2. Cyclo-stationary-based detection;
3. Radio identification-based detection;
4. Waveform-based detection.

Spectrum sharing

CR users transmit frequency band information with neighboring CRs once the frequency band has been determined. CR users must coordinate their actions because wireless channels are shared. Architecture

(centralized and distributed), spectrum allocation behavior (cooperative and non-cooperative), spectrum access technique (overlay spectrum sharing and underlay spectrum sharing), and scope (overlay spectrum sharing and underlay spectrum sharing; has two types: intra-network and inter-network spectrum sharing) are the four categories of spectrum sharing.³⁴

Spectrum management

It allocates the best spectrum available for the user’s communication while minimizing interference to other (primary) users. To meet quality-of-service criteria, CR optimizes the spectral band. As a result, spectral management skills are required for CR.³⁵

The key issue with implementing spectrum management capabilities is that they are sophisticated and multi-user at the same time. Because, it requires a variety of technical and regulatory restrictions to detect other users’ thresholds. This could indicate that national law specifies radio spectrum access laws and restrictions.

Spectrum mobility

When PU requires its licensed band for this function, CR users shift/move to the other licensed band to ensure a smooth connection. In CRNs, spectrum mobility strives to deliver a smooth and quick transition out of a spectrum handoff with minimal performance loss. A network protocol may require updates to the operational settings when a CR user alters its frequency of operation.

CR with various emerging technologies

Fuzzy logic

The FL sets of hypotheses were declared by Lotfi in 1965; A Zadeh¹⁷ used numerical and observational models to tackle and demonstrate vulnerability, equivocalness, imprecision, and dubiousness. In FL, factors are not restricted to just two qualities (True or False), as in old style and fresh sets.⁸

Genetic algorithms

Algorithms with genetic elements, Friedberg (1958), endeavored to prompt learning by altering short FORTRAN software engineers, which prompted the improvement of GAs. Therefore, a software engineer with elite execution for a specific straightforward undertaking can be produced by making a suitable series of small changes to a machine code developer.³⁶

Artificial neural network

A sort of computer network is ANNs. ANNs were invented by Warren McCulloch and Walter Pitts in 1943, with inspiration from the focused sensory system. The ANN, like the natural NN, will be made up of hubs, also known as neurons or handling components, that are connected to the network's framework. ANNs gather data from nearby nodes and generate outcomes based on their weights and beginning capabilities. The variable loads could be used to reflect the strength of neuronal connections. The loads should be modified until the network's result matches the ideal outcome to complete the learning system. To enable the CR to learn from its surroundings and make decisions to improve the correspondence framework's nature of administration, counterfeit ANNs were utilized.^{37,38}

Table 1 also provides a detailed overview of new technologies in CRNs, such as AI, FL, and GAs.

Applications of CRN

To ensure the end clients' administration, the modern wireless network must fulfill the constantly increasing data transfer capacity requirements (QoS). The board can increase data transfer capacity beyond its usual cut-off points with a productive electromagnetic range and CR innovation.³⁹ The CRNs' innovative range the board considers unlicensed (intellectual) clients using the officeholder range band without interfering with occupier clients.⁴⁰ The CRN is a smart and adaptable remote communication system in which CR devices learn from environmental elements and complete activity-dependent learning. CR devices are particularly intelligent in that they can gradually select transporter frequency, transfer speed, transmission rate, transmission power, and other parameters. As a result of CR advances, several new CR network applications are being created.⁴¹ These outstanding issue centers presented cutting-edge research findings on the use of CRN. It is founded on a thought-provoking and insightful discussion of recent developments in the application of CRNs and future directions. The application of CRSN's in-body sensors is discussed in this article by H Serrano Han et al.⁴² It depicts a lovely home and sketches out an antique dwelling framework using the CRSN. This post suggests CRSN research headlines for older, more experienced property managers. As a result, the focus of the article is on embracing CRSN progress to adapt to dense sensors and heterogeneous network conditions.

Public security communication

A CRN is used for public safety communications using white space. CR is a type of wireless communication

that allows a transceiver to distinguish between active and inactive communication channels. To do so, avoid the busy channel and immediately move to the open one.⁴³ It has no effect on the interference of licensed users.

DSA

In the investigation of DSA networks, CR is quite useful. It can quickly identify the band when the PUs are displayed. A wireless sensor network with hundreds of sensor nodes spread across the sensing region and a few meters between neighboring nodes uses CR.⁴⁴ It also looks at how sensor/network groups are evolving in relation to prior types and how sensor network developments are being combined with building pushes.

Sensor virtualization module

"Estimation of main channel activity statistics in CR based on imperfect spectrum sensing," by Pandit and Singh⁴⁵ describes the Sensor Virtualization Module (SVM). Few IoT assets use apps because of the typical smokestack programming architecture, in which suppliers supply support programming from start to completion. Alabama et al.⁴⁶ describe a lightweight and powerful solution that correctly detects channel choice correspondence in their work "SVM: Virtualizing IoT Devices on Mobile Smart Phones for Effective Sensor Data Management." Another organizational paradigm depicted in this article is cognitive body sensor networks (CBSNs). In this network of local networks, consistent accessibility is crucial and must be ensured. Setbacks in the network during an emergency could prevent a patient from receiving timely clinical care, which could be fatal.

Detection of primary user emulation attack

The proposed approach principal user emulation attack (PUEA) relies on cryptographic local persons that require a little amount of memory and low energy consumption, making it more suitable for devices with limited resources. It ensures the safety of control data provided by CBSN sensors to select a certain channel. L Liu et al. present an energy-efficient layered video multimedia (LVM) transmission over an Orthogonal Frequency Division Multiplexing (OFDM)-based CR framework for video networks using "A new spectrum scheduling technique with ant colony optimization algorithm." van Otterlo M and Wiering M, Yau K-L, and Barve S and Kulkarni P offer a power fragment computation using inadequate programming and a sub-propensity technique based on energy utility (EU). When the unit transmit power is consumed, the main show meter EU is supplied to measure the refined

Table 1. Detailed analysis of emerging technologies in CR.

Parameters	AI	FL	GAs
Description	An NN is an information processing system that is based on how biological nerve systems, such as the brain, process data. • The algorithm reflects natural selection, based on Darwin's theory of survival of the fittest. • It naturally represents the process of selecting the fittest part.	FL is a computing method based on "degrees of truth" instead of the standard "true or false" (1 or 0) binary logic used by modern computers. • Define linguistic terminology and variables (start). • For them, create membership functions (start). • Create a rule knowledge based (start). • Using membership functions, convert crisp data into fuzzy datasets (fuzzification). • In the rule basis, evaluate the rules (inference engine). • Combine the outcomes of each rule (inference engine). Convert the output data into values that are not fuzzy. It is being less complex.	A GA is a technique toward addressing limited and unregulated optimization issues that uses a natural selection process similar to biological evolution. • The program chooses a fitness function based on the starting population. • The fitness function aids in the generation of an optimal or near-optimal solution by the algorithm. • The population is maintained and evolved via the algorithm's screening, crossover, and mutation procedures. • It creates numerous populations until the optimization restrictions are met. Depends on fitness/ objective function. Minimum elapse time. This is an artificially intelligent and adaptable system. • Utilized in a variety of optimization issues. • Framework utilizes more contentions, as this string develops longer. • It is resolving multi-objective issues. It can retrieve new patterns for network classification • It can empower mindfulness handling, • Navigation and learning, the traveling salesperson problem (TSP), • GPS system
Basis of system algorithms			
Complexity	It being more complex than FL.		
Processing time	Little		
Flexibility	This system is difficult to modify.		
Utilization	• It assists in performing the predictions, makes decisions to improve the correspondence framework's nature of administration and to enable the CR to learn from its surroundings.	• Performing pattern recognition. • Having the potential to direct machines and consumer goods. • May not provide exact logic, but it does provide adequate reasoning. • Used by engineers to deal with ambiguity. It may necessitate extensive verification and testing. • In information transmission task, • obstacle and power the board, range openness examination methods, and resource segment.	
System's training	It trains itself by learning from dataset.		
Applications	• Pattern recognition; likewise, spam detection in email and cancer detection in the human body. • Forecast, that is, weather forecasting and stock market prediction		

(continued)

Table 1. Continued

Parameters	AI	FL	GAs
Working factors	- Quality and the quantity of the data provided. - Initial weights. - Cumulative weight adjustment versus incremental updating. - The steepness of the activation function λ - Learning constant η and momentum method.	- Recombination operator is the real factor behind the whole system. - Crossover probability specifies how frequently crossover will be conducted. - Mutation probability indicates how frequently chromosomal segments will be modified.	- The middle recurrence, - transmission power, and regulation sort. - They utilize historical data to take the search to the best performing region within the solution space. - Employ the concept of genetics and natural selections to provide solutions to problems.
Knowledge source	Sample sets	Human experts	Chromosomes set
Learning mechanism	Adjusting weights	Induction	Optimization technique
Reasoning mechanism	Parallel computation	Heuristic search	Metaheuristic
History	Warren McCulloch and Walter Pitts concocted ANNs in 1943, drawing motivation from the focal sensory system.	In the 1960s, Lotfi Zadeh of the University of California in Berkeley proposed the concept of FL	Algorithms with genetic elements, Friedberg (1958), led research to the improvement of GAs.

notion of reproduced video at all partners.⁵⁰ The goal is to expand the EU as a whole while also developing the intersession/interlayer subcarriers that come and go in terms of control responsibilities. To achieve the fair, it does a subcarrier job for the base layer and updates layers with appropriate estimations, resulting in an optimal power task evaluation for constructing the future EU using fragmented programming.

FL with cross-layer approach

In this research by Arunthavanathan et al.,⁵¹ the creators offer a dynamic frequency allocation, unusually termed CRN. The purpose of this research is that rerouting is costly to the amount of energy, time, and throughput. This approach is more intelligent to choose a path, needing smaller channel trading. This research examines how tactical things may perform over dynamic frequencies out the hypothesis for directly trading the degree of circumstances and present an intriguing course confirmation framework to coordinate the constant channel exchanging. Since unneeded commitment on a certain place point causes network dispersing and starts to continue rerouting, the suggested demonstrate circles the network lifespan is prolonged by the organizational overheads among academic consumers in the network. This displays the join's capacity to mind and gets at data using a cross-layer method.^{52–54}

Challenges, limitations, and strength

Fuzzy logics

Challenges. Fuzzy controllers provide the benefit of allowing specialists to apply their qualitative understanding of operations. Experts, however, find designing fuzzy controllers using their heuristic approach difficult.

Limitations. The accuracy of these systems is harmed since they rely on incorrect data and inputs. There is no one-size-fits-all approach to solve the problem with FL.⁵⁵ Because the results are frequently erroneous, it is not commonly recognized.

Strength. FL enables enhanced and more efficient machine control while also lowering costs. Although FL has been criticized for being imprecise, the conclusions are acceptable, especially when dealing with faulty inputs. FL is essential for forecasting future events.

FL is adaptable, taking into account both natural knowledge and direct computation, execution, and translation.⁵⁶ As a result, the main advantage of FL for

continuous applications such as CR is its simplicity and adaptability. When the information is ambiguous, loud, or insufficient, FL can be used to detect patterns.

Genetic algorithms

Challenges. GAs are a randomized heuristic inquiry strategy in which the population contains rival arrangements established through transformation and intersection. Moreover, GA is vital and uncomplicated because the upsides of the well-being aim capability are used for streamlining. Besides, GA may not merge to a worldwide ideal, notably in populaces with numerous individuals and execution pointers. GAs require earlier information, which is based on learning and inferred wellness functions: new rules are developed based on the training instances and trends found in prior search phrases.^{22,57,58} The challenge must be set up in such a manner that future generations are encouraged to pick better genes, and the parameters must be chosen to reflect the fitness function. Since the well-being evaluation is computationally demanded, GAs are sluggish. Because they depend on the examined issue, determining chromosomal representation of parameters, domain, and range is problematic.

Limitations. Because development based on wellness capacities cannot provide consistent enhancement reaction times, the employment of continuous GAs is limited. FL is used to show frameworks that are difficult to show due to ambiguous quality boundaries.

Strength. GA tackles a multi-objective enhancement issue and arranges the CR progressively considering the changing remote climate. GAs are faster and consume less memory while looking through a vast region.

Artificial neural network

Challenges. The ANN is based on the natural sensory system and is used to complete the learning system, find new examples, group them, and improve the dynamic interaction. To assemble NNs, a few tests are required, reducing the complexity of the arrangement.⁵⁹ ANNs could be integrated with Case Based Reasoning (CBR) and GA during the preparation stage. Because they are required, ANNs are called administered learning.

Limitations. Experimental danger minimization is the premise of ANNs. When many needs must be accomplished at the same time, the technique provides for

answers. NNs may take a long time to prepare depending on the size of the network.

Strength. They provide the ability to respond to modest climate changes and particular information about the option. ANNs are extremely adaptable and may portray a wide range of abilities. In any case, they are not sequential or deterministic. Furthermore, once the NN has been properly created, it will desire to work in forward spread mode as an insightful device on various types of data. The enlarged model for forward spread run information would then be used for further analysis and comprehension. Furthermore, an NN can be over-prepared, indicating that the network is not ready. For ANNs, infinite recursion and organized portrayals can be excessive.⁶⁰

Conclusion

This investigation depicted a study of the use of AI methods to CRNs. To arrive at an acceptable conclusion, a fuzzy inference system is based on the experiences of a group of network specialists. Data-driven models are used to supplement physically based models. It has the potential to displace physically based models. When physically based models are not possible to build due to a lack of process data, AI techniques can be used to generate a model of the process. The literary assessment of cutting-edge achievements in applying AI processes to CR is offered and organized into the fundamental artificial reasoning techniques listed below. This study looked into CR assignments and roadblocks, and the critical evaluation and challenges of using the learning technique in CRNs. Finally, the article presented a variety of perspectives and approaches to dealing with the use of learning in CRNs.

Future scope

We looked at a number of AI strategies that could be applied in CRNs. These algorithms have a lot of potential for use in CRNs. We have only recently begun investigating the use of machine learning in the CRN. There will be many more advancements in this sector. NNs must be properly built for the challenge; otherwise, performance may suffer; yet, when utilized correctly, they are a very effective tool for solving problems.^{61,62} There are more forms of NNs than ANNs, such as convolutional neural networks (CNN), recurrent neural networks (RNN), and long short-term memory (LSTM). It will be fascinating to investigate

their potential applications in CR. Combining various NNs in a CRN can also be fascinating.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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